

COMP1682 | Final Year Project

Project Proposal

**Spacic — A Synchronous Social-Audio Music Platform with
Persistent Creator-Owned Listening Rooms**

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1. Introduction

Music streaming has become the dominant commercial model for recorded music. IFPI reports global recorded-music revenues of US\$31.7 bn in 2025, of which streaming accounted for 69.6 percent and paid subscriptions for 52.4 percent. Yet despite this commercial maturity, mainstream platforms still treat listening as an individual, algorithmically mediated activity. Spotify Advertising's The Togetherness Effect shows that 74 percent of digital-audio consumers co-listen at least weekly (86 percent among Gen Z). The gap is not access — it is the social and participatory layer around access.

I want to take this project on because I have spent two years on the back end of social apps and have repeatedly noticed the same pattern: synchronous listening is reproduced with workarounds (Discord screen-share, FaceTime, sending links). That is product-market demand surfacing through friction. Building a purpose-designed artefact for synchronous social listening is therefore both technically interesting (real-time room state, sync-checkpoint algorithms, payment safety) and contextually grounded.

This direction is reinforced by two strands of academic theory and one geographic shift. First, social presence theory (Short, Williams and Christie, 1976) argues that media which let participants feel co-located with others produce stronger affective bonds than asynchronous channels — exactly the property a synchronous listening room offers and a personal playlist does not. Second, parasocial interaction theory (Horton and Wohl, 1956), updated for the platform era by Baym (2018) in *Playing to the Crowd*, frames the creator–audience relationship as a one-sided but emotionally meaningful bond that is sustained by repeated co-presence — again, what a persistent creator-owned room is designed to provide. Third, the geographic centre of streaming growth has moved: Music Ally reports that 78.4 percent of new music subscribers in 2024 came from outside North America and Europe (Music Ally, 2025); DataReportal reports Vietnam at 84.2 percent internet penetration with 137 million cellular connections (DataReportal, 2026). A platform that takes synchronous social listening seriously is therefore credible in markets that legacy desktop-first platforms underweight.

The domain of the project is digital music platforms, specifically the intersection of real-time audio synchronisation, the creator economy, and community-led discovery. The artefact, Spacic, will be a full-stack web platform on which any creator owns one persistent listening room. When the creator goes live, listeners hear the same audio at the same time and can chat, react, vote-skip, favourite the room and donate to the creator. Each room becomes a recurring social space rather than a one-off listening party.

The project will be implemented as a distributed, multi-service architecture. Likely technologies are React with TypeScript on the frontend; Node.js with Express and Socket.IO on the backend; MongoDB for durable persistence and Redis for transient room state; a separate FastAPI recommendation microservice with collaborative filtering; Clerk for authentication; Stripe and Stripe Connect for billing and creator payouts; AWS S3 for audio storage; Docker, Nginx and Jenkins for deployment and continuous integration. Methodologically the work will follow an Agile-inspired iterative approach with six two-to-

three-week sprints, MoSCoW prioritisation, and Jira-tracked user stories with acceptance criteria.

Key phrases:

synchronous social listening, persistent creator rooms, real-time audio synchronisation, co-listening, Socket.IO, Redis, MongoDB, Stripe Connect, Clerk, recommendation microservice, ALS collaborative filtering, MoSCoW, Agile / Scrum, Jira, CI/CD, creator economy, UK GDPR, ACM Code of Ethics, BCS Code of Conduct.

2. Aim

The aim of this project is to design, implement, evaluate and critically reflect on Spacic, a full-stack synchronous social-audio music platform built around persistent creator-owned listening rooms. The goal is to demonstrate that a single web artefact can integrate (a) real-time co-listening within ± 150 ms drift between participants, (b) creator-owned monetisation through an in-app wallet, donations and subscription tiers, (c) social-graph features that reduce cold-start friction, and (d) a recommendation microservice that supports discovery, all delivered to an appropriate level of professional software-engineering competence including containerised deployment, CI quality gates and security hardening.

The direction taken to reach this goal is a research-informed, iterative, prototype-driven Agile process organised into six sprints. Each sprint will deliver a vertical slice of working software, traced through Jira user stories with acceptance criteria, peer-reviewed pull requests on GitHub and automated quality gates in Jenkins. The artefact will then be evaluated against the requirement set through automated unit tests, system-level checks and user acceptance testing, and the work will be critically reflected upon in a 10,000–12,000-word final report.

Success Criteria

Five concrete, measurable criteria define a successful artefact: (a) at least one hundred and twenty passing automated unit tests run inside a hermetic CI image; (b) at least fifteen successful user-acceptance test scenarios spanning Listener, Creator and Administrator journeys; (c) synchronised playback within a ± 150 ms drift target measured against the creator's clock; (d) idempotent webhook handling for both Stripe (HMAC-signed) and Clerk (svix-signed) such that duplicate or out-of-order delivery never produces double side effects; and (e) a deployed artefact reachable over TLS via a documented production URL by 17 April 2026. These criteria are tracked through the Jira backlog and reviewed at every sprint boundary.

Approach and Methodology

The project follows an iterative, incremental and prototype-driven development approach, conducted within a Design Science Research frame (Hevner, March, Park and Ram, 2004) which legitimises a build-and-evaluate study where the contribution is an instantiated artefact. The lifecycle follows Scrum-inspired sprint structure adapted for a single-developer FYP context — sprint planning and review without team-style ceremonies, MoSCoW prioritisation,

story-pointed backlog tracked in Jira (project key SPC). Practice methodologies layered over the lifecycle include evolutionary prototyping (each sprint extends the previous artefact rather than discarding it), test-informed development (Vitest assertions encoded as acceptance criteria in test-file docstrings rather than strict TDD), and risk-driven planning drawn from the Spiral model (Boehm, 1988) where the most uncertain elements are addressed in the earliest sprint that can support them. Research methodology is dominantly secondary — literature review across industry reports and academic theory, plus comparative application analysis — supplemented by a fifteen-scenario user-acceptance pass with at least three external participants. No primary user research was undertaken in scope, owing to the ethics-approval timeline; this constraint is acknowledged honestly throughout the report.

3. Objectives

The aim is decomposed into seven SMART objectives, each scoped to be Specific, Measurable, Achievable, Relevant and Time-bounded. Each objective is supported by a set of Activities and produces concrete Deliverables.

3.1 Research the domain and complete a literature review

Establish a defensible academic and market context for the platform by reviewing streaming-economy literature, co-listening research, creator-economy data and comparable products. Completion criterion: an annotated literature review of at least 10 sources by 26 January 2026.

3.1.1 Activities

- Search and screen primary, secondary and tertiary sources (IFPI, MIDiA via Music Ally, Luminate, DataReportal, Spotify Advertising, RMIT Vietnam, Grand View Research).
- Read and annotate the most relevant 10–15 sources.
- Compare existing platforms (Stationhead, Discord) as similar-application analyses.
- Synthesise findings into a coherent argument that justifies a synchronous, room-based, creator-led platform.

3.1.2 Deliverables

- Annotated bibliography of ≥ 10 sources.
- Literature review section ($\approx 3,000$ words) ready to drop into the final report.
- Comparison tables for at least two similar applications.

3.2 Specify functional and non-functional requirements

Translate the literature findings and stakeholder analysis into a complete, traceable requirement set, prioritised using MoSCoW and tracked as Jira user stories with acceptance criteria. Completion criterion: requirement specification and Jira backlog signed off by 5 February 2026.

3.2.1 Activities

- Identify three stakeholder groups (Listener, Creator, Administrator).

- Derive 15 functional and 8 non-functional requirements.
- Apply MoSCoW prioritisation to every requirement.
- Open a Jira project (key: SPC) and create a user story per requirement with acceptance criteria.
- Map requirements to prototype phases for iterative-incremental development.

3.2.2 Deliverables

- Requirement specification document (FR-1...FR-15, NFR-1...NFR-8).
- MoSCoW table classifying every requirement.
- Jira backlog of ≥ 25 stories with story points.
- Outline prototyping plan mapping requirements to four prototype rings.

3.3 Design the system architecture, data model and user interface

Produce architecture, UML, data and deployment artefacts that demonstrate professional software-design judgement. Completion criterion: design package reviewed and frozen for first prototype by 12 February 2026.

3.3.1 Activities

- Choose a multi-service architecture and justify against alternatives.
- Produce a class diagram, ER diagram, deployment diagram and three sequence diagrams.
- Sketch listener and creator UX journeys (onboarding, room, studio, wallet).
- Document MongoDB schema, Redis key model and Socket.IO event taxonomy.

3.3.2 Deliverables

- Architecture, use-case, class, ER, deployment and sequence diagrams.
- UX wireframes for the listener and creator journeys.
- Data-model and Socket.IO event reference document.

3.4 Implement the platform across six iterative sprints

Develop the artefact through six two-to-three-week sprints, each closing with a merged GitHub pull request that passes the Jenkins quality gate. Completion criterion: a deployed, integration-tested artefact behind Nginx by 17 April 2026.

3.4.1 Activities

- Sprint 1 — Identity, Clerk webhooks and RBAC admin (SPC-1, SPC-11, SPC-19).
- Sprint 2 — Persistent rooms, sync-checkpoint playback, interactive chat (SPC-13, SPC-16).
- Sprint 3 — Stripe wallet, donations with escrow, subscription tiers, security hardening (SPC-20, SPC-14, SPC-15, SPC-79, SPC-80).
- Sprint 4 — Friends activity, friend requests, group invites, friend search, analytics, favourites (SPC-18, SPC-55, SPC-56, SPC-57, SPC-81, SPC-82).

- Sprint 5 — FastAPI recommendation microservice, Stripe Connect payouts, two-currency minigames (SPC-86, SPC-87, SPC-88).
- Sprint 6 — Hardening, Constellation visualisation, recap, presigned URLs, Docker hardening, CI lint-blocking gate (SPC-89–95).

3.4.2 Deliverables

- Six merged sprint pull requests on github.com/acedss/spacic.
- Containerised multi-service deployment via docker-compose.
- Jenkinsfile with parallel hermetic quality gates inside node:22-alpine.
- Integration-tested artefact behind Nginx with TLS.

3.5 Manage commercial and technical risk through a tracked register

Identify, monitor and mitigate the principal risks of a real-time, monetised, multi-service platform. Completion criterion: a maintained risk register reviewed at every sprint planning ceremony.

3.5.1 Activities

- Maintain a risk register covering Stripe webhook ordering, Socket.IO session recovery, MongoDB startup race, Redis retry behaviour, music-licensing exposure and adoption risk.
- Apply mitigations: idempotency keys, `recoverSessionFromDB()`, `awaited connectDB()`, bounded Redis retries, scope deferral for licensing, MoSCoW deferral for adoption.

3.5.2 Deliverables

- Risk register (initial version included in section 6 of this proposal).
- Sprint-by-sprint risk review notes.

3.6 Verify and validate the artefact through automated and user acceptance testing

Establish that the artefact behaves correctly under expected and adverse conditions and is acceptable to representative end users. Completion criterion: ≥ 120 passing unit tests and ≥ 15 successful UAT scenarios by 24 April 2026.

3.6.1 Activities

- Write Vitest unit tests for the highest-risk services (vote threshold, wallet, minigame, socket-manager, room features, auth middleware, Stripe webhook, room service, subscription service, tier features).
- Run the suite inside the Jenkins quality gate on every push.
- Recruit at least three representative participants and run a 15-scenario UAT covering listener, creator and admin journeys.
- Capture and triage any usability friction surfaced by participants.

3.6.2 Deliverables

- ≥ 120 passing unit tests in 10 suite files.

- UAT scenario sheet with per-scenario outcomes and participant initials.
- Defect / friction log with resolution notes.

3.7 Critically reflect, document and submit the final report

Produce a 10,000–12,000-word formal dissertation report that contextualises the work, justifies decisions, evaluates the outcome and reflects critically on lessons learned. Completion criterion: report submitted by 28 April 2026.

3.7.1 Activities

- Draft and revise the eight required sections (Abstract, Introduction, Literature Review, Requirement Analysis, Software Design, Software Implementation, Evaluation and Conclusion, Appendix).
- Embed all design diagrams and Jira/GitHub evidence.
- Self-review against the COMP1682 learning-outcome rubric (B–F).
- Iterate with supervisor feedback.

3.7.2 Deliverables

- Final report Report.docx (10,000–12,000 words).
- Embedded artefact evidence (test output, UAT, screenshots).
- References list in Harvard style.

4. Plan

4.1 Work Breakdown Structure (WBS)

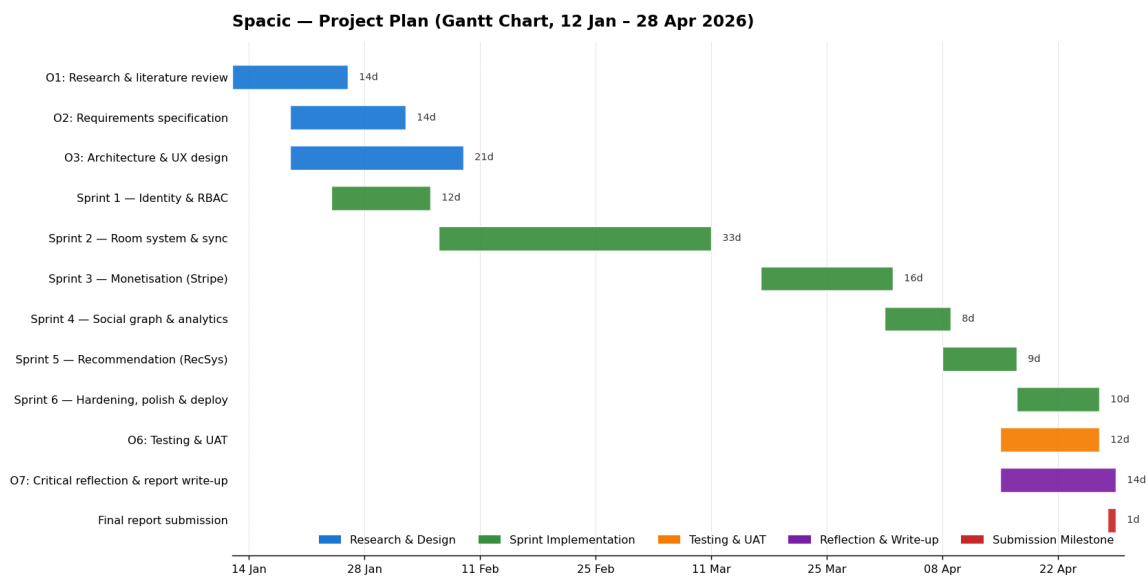
The WBS decomposes the project into three layers. Layer 0 is the project itself, Layer 1 lists the seven objectives, and Layer 2 lists the activities under each objective. Activities also map to specific Jira tickets in the SPC project.

Layer 0	Layer 1 (Objective)	Layer 2 (Activity)
Spacic project	O1: Research & literature review	A1.1 Source search & screening
		A1.2 Annotated bibliography
		A1.3 Similar-application comparison
	O2: Requirements specification	A2.1 Stakeholder analysis
		A2.2 Functional + non-functional requirements
		A2.3 MoSCoW prioritisation
		A2.4 Jira backlog creation
	O3: Architecture & UX design	A3.1 Architecture & deployment diagrams
		A3.2 Class, ER, sequence diagrams

		A3.3 UX wireframes (listener + creator)
	O4: Implementation (six sprints)	A4.1 Sprint 1 Identity & RBAC
		A4.2 Sprint 2 Rooms & Sync
		A4.3 Sprint 3 Monetisation
		A4.4 Sprint 4 Social graph & analytics
		A4.5 Sprint 5 Recommendation microservice
		A4.6 Sprint 6 Hardening & polish
	O5: Risk management	A5.1 Risk register & sprint reviews
	O6: Testing & UAT	A6.1 Vitest unit suites
		A6.2 UAT with 3 participants, 15 scenarios
		A6.3 Defect triage & fix
	O7: Reflection & write-up	A7.1 Draft 8 report sections
		A7.2 Embed diagrams & evidence
		A7.3 Final review & submission

4.2 Gantt Chart

The Gantt chart below visualises the seven objectives and the six implementation sprints across the project window of 12 January – 28 April 2026. Sprints overlap with Testing in the final fortnight to allow defect triage before the 28 April submission milestone.



4.3 Sprint Summary

Sprint	Window	Focus	Story Points
1	24 Jan – 5 Feb	Identity, Clerk webhooks, RBAC admin	16
2	6 Feb – 11 Mar	Rooms, real-time sync, chat	21
3	17 Mar – 1 Apr	Monetisation (wallet, donation, subs)	19
4	1 Apr – 8 Apr	Friends, analytics, favourites	15
5	8 Apr – 17 Apr	Recommendation, Stripe Connect, minigames	24
6	17 Apr – 26 Apr	Hardening, polish, deploy	21
Total	≈17 weeks	—	116

4.4 Critical Path Analysis

Of the seven objectives, five sit on the critical path that determines whether the 28 April 2026 submission deadline is achievable: O1 Literature Review (gates objective formulation), O2 Requirements (gates implementation), O3 Design (gates Sprint 1), O4 Implementation Sprints 1, 2, 3, 5 and 6 (each gates the next or the deployment), and O7 Reflection and Submission. The dependency chain runs Sprint 1 Identity → Sprint 2 Persistent Rooms → Sprint 3 Wallet → Sprint 5 Recommendation → Sprint 6 Hardening, because each sprint after Sprint 1 either consumes the previous sprint's outputs (rooms depend on identity, wallet attaches to rooms, recommendation works against the room corpus) or feeds the deployment gate at the end of Sprint 6. Off-path elements that can absorb a slip without missing the deadline are Sprint 4 Social Graph and the Could-Have items in Sprint 6 (Constellation visualisation, sparklines, recap). The off-path items will be deferred from sprint scope first if any earlier sprint runs over budget; this is the principal mechanism by which the schedule remains feasible.

5. Initial Risk Analysis

A risk register is maintained from the start of the project and reviewed at every sprint planning ceremony. The initial register identifies the principal commercial and technical risks together with their likelihood, impact and mitigation.

#	Risk	Likelihood	Impact	Mitigation
R1	Stripe webhook ordering not deterministic; downgrade-before-upgrade race	Med	High	Make every webhook handler idempotent and status-aware
R2	Socket.IO session lost after server restart	High	Med	recoverSessionFromDB() rebuilds Redis from MongoDB on first room:join
R3	MongoDB connection race on container start	Med	High	await connectDB() before httpServer.listen()
R4	Redis outage hangs requests with default unbounded retries	Low	High	Bounded Redis retry strategy with explicit connection timeout

R5	Music-licensing exposure for any catalogue distribution	High	High	Scope to admin-seeded catalogue and mark public catalogue as Won't-Have
R6	Single-developer scope risk	High	Med	MoSCoW prioritisation, deferred Won't-Have items, weekly supervisor check-ins
R7	Adoption risk — users attached to existing platforms	High	Low	Out of scope for the artefact; surfaced as a commercial risk in the report
R8	CI/CD environment drift between dev laptop and production	Med	Med	Hermetic quality gates inside node:22-alpine in Jenkins
R9	Real-time drift between listeners exceeds usability tolerance	Med	High	sync_checkpoint every 2 s; ± 150 ms drift target validated by UAT
R10	Personal-data exposure under UK GDPR	Low	High	Delegate identity to Clerk; cookies + JWT; rate-limited routes

6. Legal, Social, Ethical and Professional Issues (LSEPi)

6.1 Legal

The platform processes personal data (profile, activity, transactions) and is therefore in scope for the UK GDPR and the Data Protection Act 2018. Music playback raises questions under the Copyright, Designs and Patents Act 1988. Live social interaction invokes the spirit of the Computer Misuse Act 1990. Mitigations include using Clerk for identity (rather than storing passwords ourselves), using Stripe for card data (reducing PCI scope), restricting the audio catalogue to admin-seeded content, and hardening the backend with Helmet, rate limiting and CORS allow-listing.

Specific UK-GDPR articles in scope include Article 13 (information to be provided where personal data are collected from the data subject), Article 15 (right of access), Article 17 (right to erasure), and Article 20 (right to data portability). The Stripe Connect integration delegates card-data handling to a PCI-DSS Level 1 Service Provider, materially reducing the platform's own PCI scope to the SAQ-A category for redirect-based checkouts. Where webhook handlers persist limited transactional metadata into MongoDB Atlas, that metadata is encrypted at rest by Atlas (default AES-256) and in transit by TLS 1.2 or higher; Atlas's IP-allow-listing further restricts ingress to the application's known network egress. These design choices map to a defensible posture under UK GDPR obligations and the corresponding ICO guidance, even though the artefact's user base during the FYP window is limited to test participants and supervisors.

6.2 Social and Ethical

Synchronous social audio can support belonging, creator-fan relationships and niche communities, but it can also intensify parasocial behaviour, donation pressure and moderation challenges. The design responds with creator-controlled feature flags rather than unrestricted social mechanics, with caps and idempotency on donations, and with explicit donor opt-out from the public leaderboard. Recommendation is one signal among several (creator follow, room favourites, social presence) so users are not subordinated to a single algorithm.

Two further ethical concerns inform the design. First, the platform's notion of "presence" — visible co-listeners, donor leaderboards, friend activity sidebars — risks intensifying social-comparison anxiety; mitigations include a per-user opt-out of public donor visibility, a per-user mute on activity notifications, and the absence of a public follower count on creator profiles. Second, the platform's monetisation surfaces (donations, stream goals, subscription tiers) sit close to the line at which financial choices are influenced by social context; mitigations include a hard minimum donation amount of one hundred credits, idempotency keys on every wallet transaction (preventing duplicate charges from rapid clicks), and clear in-UI confirmations before any transfer. The ACM Code of Ethics clauses 1.1 (contribute to society and human well-being), 1.2 (avoid harm) and 2.5 (give comprehensive and thorough evaluations of computer systems) directly inform these design choices.

6.3 Professional

The project is conducted in line with the ACM Code of Ethics and the BCS Code of Conduct, both of which emphasise privacy, avoidance of harm, public interest, competence and honesty. Where a third-party service handles a high-risk responsibility better than a custom build (Clerk for auth, Stripe for payments), that delegation is preferred. All academic sources will be cited in Harvard style and the supervisor will be consulted before any deviation from the approved scope.

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